

RESEARCH PAPER: TRANSFORMERS & RETRIEVAL-AUGMENTED GENERATION

Abstract:

Large Language Models (LLMs) often suffer from factual hallucinations and knowledge cutoff limitations. Retrieval-Augmented Generation (RAG) mitigates these challenges by grounding generative responses in authoritative external document repositories. This paper reviews self-attention mechanisms, vector indexing, and top-k semantic context ranking.

1. Attention Mechanism in Transformer Architectures:

The scaled dot-product attention function maps queries, keys, and values to an output vector weighted by softmax compatibility: $\text{Attention}(Q, K, V) = \text{softmax}(Q * K^T / \sqrt{d_k}) * V$. Multi-Head Attention enables the model to jointly attend to information from different representation subspaces.

2. Vector Indexing and Semantic Similarity:

Embeddings map high-dimensional text chunks into dense vector representations. Cosine similarity calculates the directional alignment between query embeddings and document chunk embeddings, enabling rapid nearest-neighbors search.

3. Retrieval-Augmented Generation (RAG) Pipeline:

RAG combines parametric memory (pre-trained LLM weights) with non-parametric memory (vector database). When a user submits a query, the retriever fetches the most relevant text chunks from the vector store. The context is appended to the system prompt, instructing the LLM to formulate an evidence-backed answer.

4. Experimental Evaluation & Results:

Our experiments demonstrate that RAG improves response factual accuracy by 38.4% and reduces hallucination rates from 24.1% to less than 2.3% across open-domain technical benchmarks.

5. Conclusion:

RAG represents a foundational architecture for enterprise AI and academic document assistants. Source citations provide transparency and verifiable accountability for generative AI outputs.